4	**(a)**	Simple harmonic motion (SHM) is defined as the oscillation of a particle such that its <u>acceleration is proportional to its displacement</u> from the equilibrium position, and is always <u>directed towards the equilibrium position</u> (or opposite in direction to the displacement).		B1 B1
	(b)	**(i)**	$\omega = 2\pi f = 2\pi(3.0) = 18.85 \text{rads}^{-1}$ Maximum kinetic energy = ½ $m\ \omega^2\ x_o^2$ = ½ (0.80)(18.85) ^{2}(0.016)^2 = 0.0364 J ≈ 0.036 J	C1 A1
		(ii)	$x = -x_0 \cos \omega t$ $v = \frac{dx}{dt} = \omega x_0 \sin \omega t$ $\omega = 2\pi f = 2\pi(3.0) = 18.85 \text{rads}^{-1}$ Velocity v = (18.85)(0.016) sin(18.85(0.20)) Velocity v = -0.1773 m s^{-1} Speed v ≈ 0.18 m s^{-1} (Do not accept negative sign for final answer.)	C1 A1